

House Price Prediction Using Machine Learning: A Study on the Impact of Spatial and Environmental Features

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Abstract—This study investigates the problem of predicting house prices using machine learning techniques. The dataset used in this study contains various housing features, including structural characteristics, environmental factors, and spatial information such as city and waterfront. A complete data modelling pipeline was applied, including data preprocessing, feature engineering, and model training. Several regression models, including Linear Regression, Random Forest, and Gradient Boosting, were implemented and compared. The results show that ensemble models outperform linear models due to their ability to capture non-linear relationships. In addition, spatial and environmental features such as city and waterfront were found to have a significant impact on house prices. These findings highlight the importance of incorporating both structural and spatial features in house price prediction.

Index Terms—house price prediction, machine learning, regression, spatial features, environmental factors

I. INTRODUCTION

House price prediction is a key task in the real estate industry. Real estate plays an important role in the global economy and affects many stakeholders, including buyers, sellers, investors and policymakers [1]. This is not only because its price is deeply affected by complex geographical factors and local regulatory policies [1], but also because accurate property valuation is essential for supporting informed decision-making, ensuring market transparency, and maintaining financial stability.

However, as a highly differentiated product, the value of real estate is composed of internal characteristics (such as building quality) and external environmental attributes (such as geographical location and employment accessibility) [2]. Traditionally, house price estimation mainly relies on expert experience judgment or simplified statistical methods. However, due to the lack of market information, low transaction frequency and the complex dynamics brought about by local regulatory policies, it is difficult for these methods to capture the price micromechanism, and it is easy to produce subjective deviations and valuation inconsistency [1].

With the rapid development of data-driven technology, machine learning methods are increasingly used in house price

prediction. These methods can capture the complex and non-linear relationship between house features, thus significantly improving the predictive performance [3]. Researchers have verified a variety of algorithms, including traditional linear regression, decision tree, random forest and gradient boosting model [1], [4], [5]. Compared with traditional parameter models (such as hedonic regression), these machine learning algorithms show stronger robustness and prediction accuracy when processing large-scale and high-dimensional real estate data, especially in capturing complex nonlinear relationships.

However, although the machine learning model shows good performance, no model can always maintain the best performance in all scenarios. Different models have their own advantages and disadvantages. Linear models (such as traditional hedonic regression) are simple, computationally efficient and easy to explain, but it is difficult to capture complex nonlinear relationships in housing data [6]. Tree-based models (such as decision trees and random forests) can capture nonlinear interactions between features and process high-dimensional data [4], [7]. However, while improving the prediction accuracy, such models also face the challenge of easy overfitting and lower model interpretability than traditional linear models [7]. Neural network models can learn highly complex nonlinear patterns and interactions, but their performance is highly dependent on network structure design, and usually requires fine parameter tuning through trial and error [6]. Support vector regression (SVR) can effectively process nonlinear relationships in house price data in higher-dimensional feature space by introducing kernel functions [8]. However, this method has significant limitations in practical application: on the one hand, its predictive performance shows high sensitivity to the selection of penalty parameters C , ϵ -insensitive zone and nuclear function parameters [8]. On the other hand, as its computational complexity increases significantly with the increase of the sample size, its calculation cost is often higher than that of tree-based models (such as random forests) when processing large-scale real estate data sets.

In order to overcome the limitations of a single prediction model, the integrated learning method has been widely intro-

duced into the field of house price prediction. Among them, Stacking uses a meta-learner to integrate predictive output from heterogeneous learners (such as decision trees, random forests, gradient boosting, and neural networks) by building a multi-layer learning architecture. This mechanism can capture and correct the prediction bias of different algorithms, and significantly improve the accuracy and robustness of the overall prediction through the synergistic complementary effect [9], [10].

However, although the stacking integration method has potential, the study of systematically applying it to house price prediction tasks is still limited [11]. Existing research usually focuses on single predictive models or simple integration technologies such as Bagging and Boosting, and the systematic evaluation of the effectiveness, robustness and generalization ability of stacking integration methods on different data sets is still insufficient. Recent studies have shown that while stacking integration can improve accuracy to some extent, the improvement is often limited and computationally expensive. Therefore, the general applicability of this method in complex market environments still requires further systematic validation [12]. Furthermore, some existing studies based on stacked ensembles are still insufficient in terms of base learner contribution analysis and interpretability, which to some extent limits the application potential of such methods in real-world housing valuation [12]. Therefore, there remains a research gap in exploring whether stacking ensembles can consistently outperform individual models in real-world housing datasets.

The dataset used in this study is the King County housing dataset obtained from Kaggle, which contains 21,613 house transactions recorded between May 2014 and May 2015. The dataset includes multiple features describing each property, such as structural attributes (e.g., bedrooms, bathrooms, and living area), as well as spatial and environmental factors (e.g., waterfront and view). The target variable is the house sale price. This dataset is well-suited for modelling complex relationships in real estate pricing due to its rich feature set and inclusion of both physical and geographical characteristics.

The main research questions addressed in this study are as follows:

- How do different machine learning models perform in predicting house prices?
- Can stacking ensemble methods outperform individual models in terms of predictive accuracy, robustness, and generalization ability?

II. LITERATURE REVIEW

III. METHODOLOGY

IV. EXPERIMENTAL RESULTS AND ANALYSIS

V. DISCUSSION

VI. CONCLUSION AND FUTURE WORK

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